

SHAJI A P

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PROFESSIONAL SUMMARY

Results-driven embedded system Engineer with 8 years of experience in designing, developing, and optimizing embedded systems and software. Proficient in Embedded C/C/C++, RTOS, Microcontrollers, troubleshooting, debugging, and enhancing system performance, with a focus on reliability and efficiency.

WORK EXPERIENCES

- **Tech Mahindra Pvt Ltd (Deputed to SAMSUNG):** Senior Software Developer (January 2022 – Present)
- **CSI Interfusion India Pvt Ltd (Deputed to HUAWEI):** Software Developer (October 2018 – January 2022)
- **Doptale Interactive Pvt Ltd:** Embedded Software Engineer (October 2017 – August 2018)
- **Rabitech Technologies Pvt Ltd:** Embedded System Engineer (May 2016 – September 2017)

INTERNSHIP

- **Rabitech Technologies Pvt Ltd:** Postgraduate Intern – Embedded Systems (September 2015 – April 2016)

ACADEMIC PROFILES

- M-Tech in Embedded Systems: National Institute of Electronics and Information Technology Calicut
- B-Tech in Electronics and Communication: College of Engineering Vadakara
- Plus Two: State Board of Kerala
- SSLC: State Board of Kerala

TECHNICAL SKILLS

- ✓ Programming Languages : Embedded C, C, C++, Python
- ✓ Microcontroller : 8051, STM32 32 bit ARM based microcontroller, PSoc, Nuvoton N76E003
- ✓ Communication Protocols : SPI, I2C, UART, RS232
- ✓ Security Protocol: IKE, IPSec
- ✓ Software Tools – Code Composer studio, IAR Workbench for 8051, GCC, Code warrior for ARM, EAGLE for PCB designing, Arduino IDE, Energia IDE, Android, VS Code, Source Insight, MakeFile, STMcubeIDE, STMCubeMX
- ✓ Debugging Tool: GDB, STM Cube IDE SWD/openOCD, USB Logic analyzer.
- ✓ Other – FreeRTOS, ESP8266 Wi-Fi, CC3200Wi-Fi, CC2541 BLE, GDB, ESP32, STM32f401RE

INDUSTRIAL PROJECTS

Title : Samsung Tizen watch

Industry : Samsung R&D Institute India Bangalore Private Limited

Technology : Bluetooth, Android, Samsung Accessory protocol, Tizen OS

Industry : Contributed to the development of the Samsung Accessory Service Protocol, enabling seamless communication between smart watches and smart phones. Utilized C language to develop, debug, and optimize protocol functionalities. I diagnosed and resolved technical issues, ensuring robust performance and reliability. I effectively communicated updates and solutions to users, maintaining a high level of user satisfaction.

Title : IP Master (IKE/IPSec protocol)
Industry : HUAWEI Technology India Pvt Ltd
Technology: IKE, IPSec, Linux

Description: This project is developed in C using the Source Insight application. I gained expertise in protocols such as IKEv1, IKEv2, and IPSec while focusing on reducing code complexity and optimizing performance in alignment with Huawei's coding guidelines. My responsibilities included ensuring secure and efficient code delivery to customers, as well as debugging and resolving issues identified by the testing team related to the IPSec flow.

Title : Huawei FusionStorage
Industry : HUAWEI Technology India Pvt Ltd
Technology : Storage Protocol (NFSv3 and NFSv4.1), Linux

Description : As a member of the Storage Protocol Team, I contributed to Fusion Storage development with a focus on lower-level testing (LLT) and worked extensively on NFS version 4.1, implementing the read operation and developing LLTs for operations such as read, write, move, and copy. I designed and developed a Python-based debugging tool to analyze issue logs and streamline troubleshooting processes. My work involved leveraging C programming for protocol development and C++ for implementing and testing LLTs, ensuring efficient and reliable system performance.

Title : NEARALS
Industry : Rabitech technologies Pvt Ltd
Software/Hardware: 8051, BLE, IAR Workbench, CC254x

Description: As a Firmware and Hardware Developer for Nearals, a business optimization platform connecting customers and retailers in the offline market, I was responsible for developing the firmware, hardware design, and PCB layout for Nearals beacons. These beacons provided location-based context to the Nearals mobile app, enabling it to deliver targeted offers to users based on proximity. My contributions ensured reliable beacon functionality and seamless integration within the Nearals ecosystem.

Title : NURSE CALLING SYSTEM
Industry : Rabitech technologies Pvt Ltd
Software/Hardware: ARM Cortex M4, Wi-Fi, Node.js server, Code Composer Studio

Description: Developed a Wi-Fi-enabled Nurse Calling System to improve patient care and communication in hospitals. The system consisted of two devices: one for patients and one for the nurse station, both connected to the hospital's Wi-Fi network. Patients could call for assistance by pressing a button on their device, triggering a notification on a web dashboard at the nurse station and activating an audible alarm on the nurse station device. My role involved designing the hardware, developing firmware, and ensuring seamless system functionality to enhance reliability, usability, and response efficiency in healthcare settings.

Title : GRENDHA Interactive Game Board- Prototype
Industry : Doptale Interactive Pvt Ltd
Software/Hardware: Bluetooth, NFC, Atmega2560, Raspberry Pi, Android

Description: GRENDHA Interactive game board is a prototype of two player board game. Game contains 60 cell interactive board with a display powered by android, 4 playing pieces and a smart dice, named DICE+. Smart dice and android device connected by Bluetooth. Board and android device connected via serial. Smart dice properties like roll, tap on a face are used to control the game accordingly player makes the movement. Story and rules for the game is displays on the display.

M-TECH PROJECTS

Title : Wearable Health Monitoring Device

Software/Hardware: Bluetooth 4.1, Wi-Fi 3200, IAR, Heart rate sensor, DHT11

Designed and developed a wearable device using an ARM microcontroller with BLE and Wi-Fi capabilities to transmit patient vitals (heart rate, body and environmental temperature, humidity) and relative position to a server for real-time health monitoring and location tracking.

Title : SMART ROBOT

Software/Hardware: Arduino Uno, Bluetooth (HC-05)

Developed a robot controlled via Smartphone for 2D motion, utilizing an Arduino-based system and Bluetooth (HC-05). Integrated an Android RC application for precise control, enabling industrial applications, crack inspection, and potential use in the gaming industry.

Title : VOICE CONTROLLED ROBOT

Software/Hardware: Arduino Uno, Bluetooth (HC-05), Voice recognition Module

Developed a robot controlled by specific voice commands for 2D motion, utilizing a voice recognition module capable of storing up to 15 commands. The system detects and processes stored commands, outputting corresponding HEX values to control robot movement.

B-TECH PROJECTS

Title : MODIFIED HUFFMAN ENCODING

Software/Hardware: MatLab

Description: This project gives a brief idea about the research on the modified Huffman encoding algorithm for lossless image compression. Before doing Huffman coding, zigzag entropy coding is done. Using this coding we achieved 90% of compression without losing clarity of image.

Title : BROKEN RAILWAY TRACK DETECTION

Software/Hardware: PIC16F873, GSM, MPLAB

Description: This project is used to detect the broken railway track and to inform authority to avoid the accidents. This is a microcontroller based project. Here we use one tune generation IC and it send a particular frequency of tune send through the railway track. Any crack in rail leads failure in detection of the tune and it send alert to the preloaded number in the GSM module

JOB SKILLS

- Ability to easily acquire and learn new technologies
- Ability to work in team.
- Ability to lead a team.
- Committed to work in project deadlines and schedules.
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DECLARATION

I hereby declare that the abovementioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.

Bangalore
10-06-2025

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